

Regularization :-

Regularization is a technique used to reduce overfitting by adding a penalty term to the loss function that discourages large model weights.

→ It is mainly used when a model has high variance and performs well on training data but poorly on the unseen data.

→ Regularization increases bias slightly but significantly reduces variance, improving generalization.

→ Regularization does not make a model better on training data → it makes it better on unseen data.

When to use Regularization :-

i) Preventing Overfitting :- Regularization is most commonly used as a tool to prevent overfitting. If your model performs well on the training data but poorly on the validation or test data, it might be overfitting and regularization could help.

ii) High Dimensionality :-

→ Regularization is particularly useful when you have a high number of features compared to the number of data points.

→ In such scenarios, model tends to overfit easily and regularization can help by effectively reducing the complexity of the model.

iii) Multicollinearity :-

→ when features are highly correlated (multicollinearity), it can destabilize your model and make the model's estimates sensitive to minor changes in the model.

→ L2 regularization can help in such cases by distributing coefficients estimates among correlated features.

iv) Feature Selection :-

- If you have lot of features and you believe many of them might be irrelevant, L1 can help.
- It tends to produce sparse solutions, driving the ~~irrelevant~~ coefficients of irrelevant to zero, thus performing feature selection.

Type of Regularization

L2 Regularization
(Ridge)

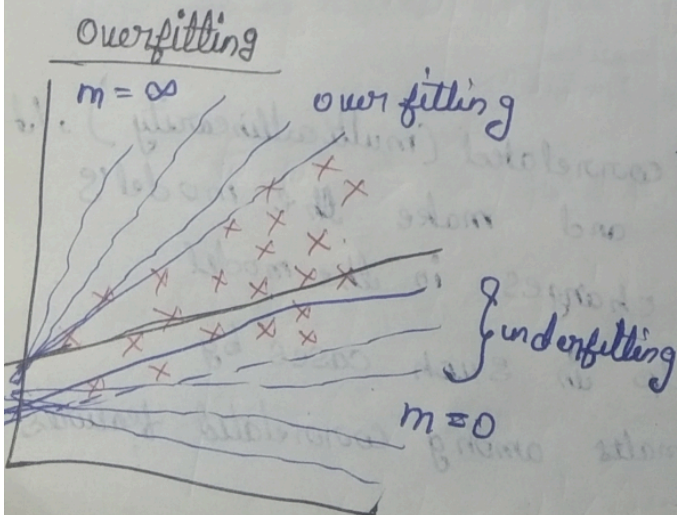
L1 Regularization
(Lasso)

Elastic Net (L1+L2)

Ridge Regression

Ridge Regression is regularized version of linear regression that add an L2 penalty to the loss function to discourage large coefficients.

- It is mainly used to reduce overfitting and handle multicollinearity.
- By shrinking coefficients, Ridge reduces variance, at the same time slightly bias increases, which improves generalization.
- Unlike Lasso, Ridge does not perform feature selection because it does not set coefficients exactly to zero.
- The regularization strength is controlled by a hyperparameter λ , typically chosen using cross-validation.

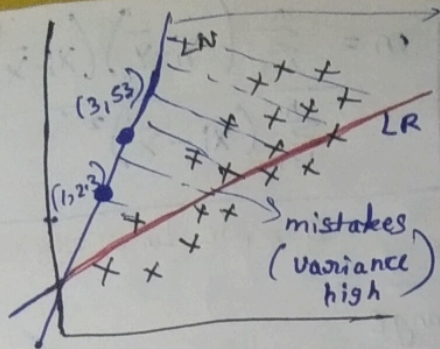


$$y = mx + b$$

$$OLS \Rightarrow \text{Minimize } \sum (y_i - \hat{y}_i)^2$$

Problem 5:-

- If features are highly correlated & data has many features.
- Coefficients become large
- Model becomes unstable



$$y_i = \frac{1.5x + 0.8}{m}$$

$$\hat{y}_i = \frac{0.9x + 1.5}{m}$$

• training data (only 2 points)

• Testing data.

$$L = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda (m^2)$$

hyperparameter

Loss LN

Loss LR

$$\lambda = 1$$

$$0 + 1(1.5)^2$$

$$\Rightarrow \boxed{2.25}$$

$$\lambda = 1$$

$$(2.3 - 0.9 - 1.5)^2$$

$$+ (5.3 - 2.7 - 1.5)^2$$

$$+ (0.9)^2$$

$$\Rightarrow (0.1)^2 + (1.1)^2 + (0.9)^2$$

$$\Rightarrow \boxed{2.03} \checkmark$$

in case i) line passing through two points so $(y_i - \hat{y}_i)$ are same. So, it will become zero.

for more features

$$\lambda (m_1^2 + m_2^2 + m_3^2 + \dots + m_n^2)$$

Mathematical Intuition:-

$$y = mx + b$$

$$L = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda m^2$$

$$b = \bar{y} - m\bar{x}$$

$\bar{x} \rightarrow x$ mean

$\bar{y} \rightarrow y$ - mean

$m \rightarrow$ slope

from linear regression, it has derived

$$\frac{\partial L}{\partial b} = 0 \rightarrow b = \bar{y} - m\bar{x}$$

$$\frac{\partial L}{\partial m} = 0 \rightarrow m$$

$$\Rightarrow L = \sum_{i=1}^n (y_i - mx_i - b)^2 + \lambda m^2 \quad [\hat{y}_i = mx_i + b]$$

$$= \sum_{i=1}^n (y_i - mx_i - \bar{y} + m\bar{x})^2 + \lambda m^2$$

$$\frac{\partial L}{\partial m} = 2 \sum_{i=1}^n (y_i - mx_i - \bar{y} + m\bar{x}) (-x_i + \bar{x}) + 2\lambda m = 0$$

$$= -2 \sum_{i=1}^n (y_i - \bar{y} - mx_i + m\bar{x}) (x_i - \bar{x}) + 2\lambda m = 0$$

$$= \lambda m - \sum_{i=1}^n [(y_i - \bar{y}) - m(x_i - \bar{x})] (x_i - \bar{x}) = 0$$

$$= \lambda m - \sum_{i=1}^n (y_i - \bar{y}) (x_i - \bar{x}) + m \sum_{i=1}^n (x_i - \bar{x})^2 = 0$$

$$= \lambda m + m \sum_{i=1}^n (x_i - \bar{x})^2 - \sum_{i=1}^n (y_i - \bar{y}) (x_i - \bar{x})$$

$$= m \Rightarrow \frac{\sum_{i=1}^n (y_i - \bar{y}) (x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2 + \lambda}$$

$$m = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2 + \lambda}$$

ridge regression

when $\lambda = 0$

there is no difference / change

when $\lambda = 1, 10, 100 \dots [0 \dots \infty]$

→ m will decrease, when $\lambda \uparrow$

→ when m ↓, it will affect b [$y = mx + b$]

[m is nothing but the weight]

$$m = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Simple linear regression

Ridge Regression for nD data:-

From $E = e^T \cdot e$ from MLP, $\hat{y} \approx X\beta$

OLS $\Rightarrow L = (y - X\beta)^T (y - X\beta) \Rightarrow (y - \hat{y})^T (y - \hat{y})$

$$X = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{bmatrix}$$

Feature matrix

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_n \end{bmatrix}$$

Coefficients

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}$$

columns

Ridge adds L2 regularization:-

$$\lambda \|\beta\|^2$$

$$L = (y - X\beta)^T (y - X\beta) + \lambda \beta^T \beta$$

$$\begin{bmatrix} \beta_0 & \beta_1 & \beta_2 & \dots & \beta_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{bmatrix}$$

$$\lambda \beta_0^2 + \lambda \beta_1^2 + \lambda \beta_2^2 + \dots + \lambda \beta_n^2$$

$$\Rightarrow \lambda [\beta_0^2 + \beta_1^2 + \dots + \beta_n^2]$$

$$\beta \cdot \beta^T \Rightarrow \lambda [\beta_0^2 + \beta_1^2 + \beta_2^2 + \dots + \beta_n^2]$$

$$(a-b)^T = a^T - b^T$$

$$L = [(y^T - (X\beta)^T) (y - X\beta) + \lambda \beta^T \beta]$$

$$= (y^T - \beta^T X^T) (y - X\beta) + \lambda \beta^T \beta$$

$$= y^T y - y^T X \beta - \beta^T X^T y + \beta^T X^T X \beta + \lambda \beta^T \beta$$

$$L = \beta^T X X \beta - \frac{2 \beta^T X^T Y}{X^T B = B} + Y^T Y + \lambda \beta^T \beta$$

→ differentiate w.r.t β

$$\frac{\partial L}{\partial \beta} = 2 X^T X \beta - 2 X^T Y + 0 + 2 \lambda \beta = 0$$

$$\Rightarrow 2 X^T X \beta - 2 X^T Y + 0 + 2 \lambda \beta = 0 \div 2$$

$$\Rightarrow X^T X \beta + \lambda \beta = X^T Y$$

$$\Rightarrow (X^T X + \lambda I) \beta = X^T Y$$

$$\begin{aligned} X^T B &= B \\ X^T b &\rightarrow b \\ X^T x &\rightarrow 2x \\ X^T B x &\rightarrow 2Bx \end{aligned}$$

$$\text{Ridge} \Rightarrow \beta = [X^T X + \lambda I]^{-1} X^T Y$$

Identity matrix

when λ value increases β value will be reduced.

$$\text{MLR} \Rightarrow \beta = (X^T X)^{-1} X^T Y$$

Ridge Regression Using Gradient Descent:-

$$L = \sum_{i=1}^n (y_i - \hat{y})^2 \Rightarrow L = (Xw - y)^T (Xw - y) + \lambda ||w||^2$$

where $w \Rightarrow$ weights / coefficients
(β)

$$L = (Xw - y)^T (Xw - y) + \lambda w^T w$$

$$X = \begin{bmatrix} 1 & X_{11} & X_{12} & \dots & X_{1n} \\ 1 & X_{21} & X_{22} & \dots & X_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{m1} & X_{m2} & \dots & X_{mn} \end{bmatrix}$$

$$y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{bmatrix} (n \times 1)$$

$$w = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} (n+1) \times 1$$

→ for $w_0, w_1, \dots, w_n$ (Parameters), need to update

$$w_0 = w_0 - \eta \frac{\partial L}{\partial w_0}, \quad w_1 = w_1 - \eta \frac{\partial L}{\partial w_1}, \quad \dots, \quad w_n = w_n - \eta \frac{\partial L}{\partial w_n}$$

$$w_{\text{new}} = w_{\text{old}} - \eta \left[\frac{\Delta L}{\Delta w} \right] \Rightarrow \text{Gradient} \Rightarrow \begin{bmatrix} \partial L / \partial w_0 \\ \partial L / \partial w_1 \\ \vdots \\ \partial L / \partial w_n \end{bmatrix}$$

$$L = (Xw - y)^T (Xw - y) + \lambda w^T w \quad [x2]$$

$$L = \frac{1}{2} (w^T X^T - y^T) (Xw - y) + \frac{1}{2} \lambda w^T w$$

$$\Rightarrow \frac{1}{2} \left[\omega^T X^T X \omega - \underbrace{\omega^T X^T y - y^T X \omega + y^T y}_{\text{Symmetric}} \right] + \frac{1}{2} \lambda \omega^T \omega$$

$$\Rightarrow \frac{1}{2} \left[\omega^T X^T X \omega - 2 \omega^T X y + y^T y \right] + \frac{1}{2} \lambda \omega^T \omega$$

$$\frac{\partial L}{\partial \omega} \Rightarrow \frac{1}{2} \left[2 X^T X \omega - 2 X^T y + 0 \right] + \frac{1}{2} [2 \lambda \omega]$$

$$\boxed{\frac{\partial L}{\partial \omega} \Rightarrow X^T X \omega - X^T y + \lambda \omega} \Rightarrow \frac{\partial L}{\partial \omega}$$

for λ in epochs:

$$\omega = \omega - \eta \frac{\partial L}{\partial \omega} \quad \text{will run epoch times}$$

Key Understanding about Ridge :-

$$\sum_{i=1}^n (y_i - \hat{y})^2 + \lambda \|\omega\|^2 \quad \left(\begin{array}{l} \text{Shrinkage Coefficients} \\ \downarrow \quad \rightarrow \lambda (\omega_1^2 + \omega_2^2 + \omega_3^2 + \dots + \omega_n^2) \\ \text{Reduce overfitting.} \end{array} \right)$$

i) How the Coefficients get affected

i) when $\lambda = 0$

- Ridge becomes normal LR
- No penalty
- Coefficients can be large
- Sensitive to multicollinearity

$$\boxed{\beta_{\text{ridge}} = \beta_{\text{OLS}}}$$

ii) Small λ (0.001, 0.01, 0.04)

- Very light penalty
- Coefficients shrink slightly
- model still close to OLS
- Variance reduces a bit

$$\boxed{\text{Small } \lambda = \text{Small shrinkage}}$$

iii) Medium λ

- Noticeable shrinkage,
- Correlated features get balanced weights
- model becomes stable

→ Bias increases, Variance decreases

$$\boxed{\beta \downarrow \text{ as } \lambda \uparrow}$$

iv) Large λ (10, 100, 1000)

- Heavy Penalty
- Coefficients move very close to zero.
- Features contribute very little
- model underfits

$$\lambda \rightarrow \infty \Rightarrow \beta \rightarrow 0$$

- Coefficients become small, but never exactly zero

ii) How very large coefficients are impacted in Ridge

- Ridge penalizes large coefficients more strongly than small ones

- Ridge adds this penalty to the loss:

$$\lambda \|\beta^2\| \Rightarrow \lambda \sum_{i=1}^n \beta_i^2$$

$$\lambda [\beta_1^2 + \beta_2^2 + \beta_3^2 + \dots + \beta_n^2]$$

- Penalty depends on β^2

- If β is large $\rightarrow \beta^2$ becomes very large

Example:- $\beta = 1 \Rightarrow \beta^2 \Rightarrow (1)^2 \Rightarrow 1$

$\beta = 100 \Rightarrow \beta^2 \Rightarrow (100)^2 \Rightarrow 10000$

In Presence of multicollinearity:-

when feature are correlated:

OLS $\rightarrow$ One Coefficient becomes very large

Ridge $\rightarrow$ breaks this dominance

$\rightarrow$ large coefficient get reduced

$\rightarrow$ weight is shared among correlated features

Why Coefficients Never become Zero:-

Ridge add this term to the loss:

$$\lambda \sum_{i=1}^n \beta_i^2$$

- Penalty is quadratic
- Curve is smooth and round
- No sharp corners

$$\beta = (X^T X + \lambda I)^{-1} X^T y$$

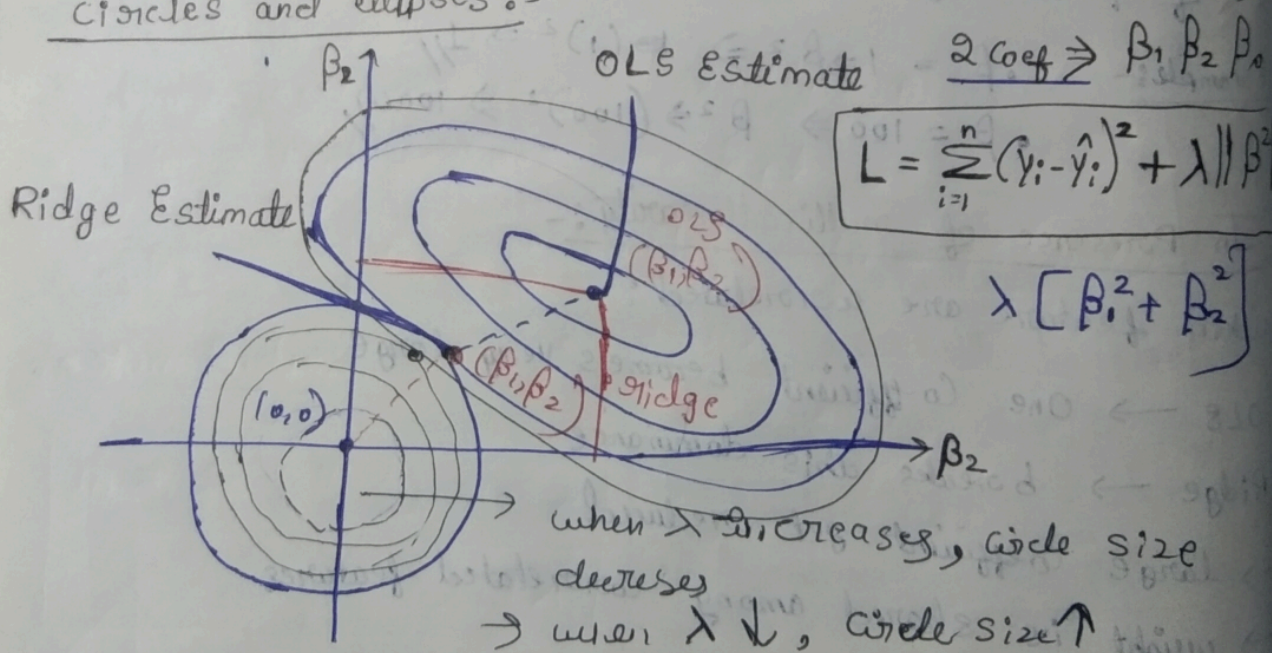
- Inverse shrinks values
- Does not set any component ~~to~~ Exactly to zero
- Shrinkage is continuous.

IN Ridge regression, coefficients never become zero because the L2 Penalty is smooth (β^2), which shrinks coefficients continuously but never forces them exactly to zero.

Why it is called Ridge:-

- Ridge regression gets its name from the shape of the "best solutions" when you add the L_2 penalty.
- They lie along a narrow, ridge-like curve on the error surface in coefficient space.

Circles and ellipses:-



- Error Contours of loss plot of OLS are drawn as ellipses around the OLS solution in the (β_1, β_2) plane.

- The ridge constraint $\beta_1^2 + \beta_2^2 \leq C$ is drawn as a circle centered at origin $(0,0)$